CLASS TEST AAHL

DATE : 27 OCT 25 **Marks - 50**

1 [Maximum marks 5]

Solve the following system of equations.

$$\log_{x+1} y = 2$$
$$\log_{y+1} x = \frac{1}{4}$$

2. [Maximum marks 5]

In the arithmetic series with n^{th} term u_n, it is given that $u_4 = 7$ and $u_9 = 22$.
Find the minimum value of n so that $u_1 + u_2 + u_3 + \ldots + u_n > 10\,000$.

3. [Maximum marks 4]

Consider the arithmetic sequence 8, 26, 44, ...

(a) Find an expression for the nth term.

(1)

(b) Write down the sum of the first n terms using sigma notation.

(1)

(c) Calculate the sum of the first 15 terms.

(2)

4. [Maximum marks 8]

The sum, S_n, of the first n terms of a geometric sequence, whose n^{th} term is u_n, is given by

$$S_n = \frac{7^n - a^n}{7^n}, \text{ where } a > 0.$$

(a) Find an expression for u_n

(2)

(b) Find the first term and common ratio of the sequence.

(4)

(c) Consider the sum to infinity of the sequence.

(i) Determine the values of a such that the sum to infinity exists.

(ii) Find the sum to infinity when it exists.

(2)

5. [Maximum marks 6]

Solve the equation $2^{2x+2} - 10 \times 2^x + 4 = 0, x \in \mathbb{R}$. [5]

6. [Maximum marks 5]

Let $\log_{10} 2 = p$ and $\log_{10} 3 = q$.

(a) Find an expression for $\log_{10} 24$ in terms of p and q.

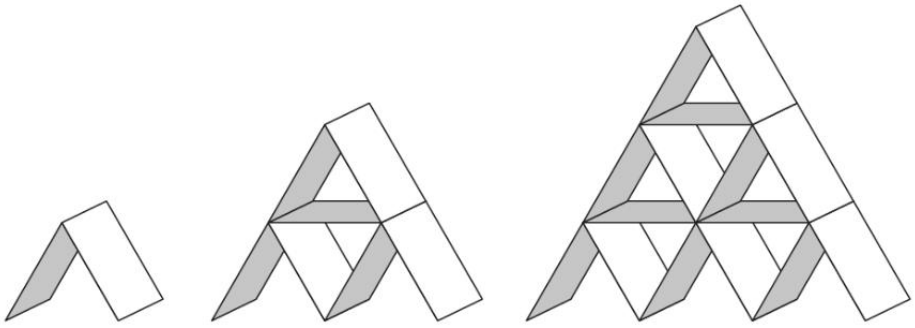
(b) Find an expression for $\log_3 8$ in terms of p and q.

7. [Maximum marks 11]

Rectangular playing cards are stacked in the shape of a pyramid with n rows, where $n \geq 1$.

Some cards are placed horizontally and some cards are stacked at an angle of 60° to the horizontal.

The following diagrams represent pyramid stacks for $n = 1$, $n = 2$ and $n = 3$.



$n = 1$ $n = 2$ $n = 3$

Let t_n represent the number of cards used to create a pyramid stack with n rows.

(a) Write down t_3. [1]

(b) Find t_4. [2]

(c) Show that $t_n = \frac{n(3n+1)}{2}$. [3]

There are 52 cards in a full pack of playing cards.

(d) A complete pyramid stack is created using playing cards taken from 14 full packs. Find the maximum number of rows in this stack. [3]

(e) A complete pyramid stack is created using playing cards taken from full packs with no cards left over. Find the minimum number of rows in this stack. [2]

8. [Maximum marks 6]

Alex purchases a car for € $30\,000$. The value of the car depreciates at $15\,\%$ per annum.

(a) Find the value of the car after ten years. Give your answer to two decimal places. [2]

Alex invests € $50\,000$ in a bank account that pays a compound interest rate of $1.5\,\%$ per month.

Inflation over the same time period was $0.8\,\%$ per month.

(b) Find the number of months required for the real value of the investment to first exceed € $55\,000$. [4]